

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 201 — Past Questions & Materials
by Renance

Course: GET 201
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

This study pack was compiled and restructured by Renance for GET 201 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 2 — GET 201 - Applied Electricity I

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

COURSE CODE: GET 20)

COURSE TITLE: Applied Electricity I

Time Allowed: 3 Hrs.

EXAMINATION: First Semester

SESSION: 2024/2025

DATE: 21/08/2025

INSTRUCTIONS:

1. Answer at least one question from each section

2. Do not answer more than two questions from each section

3. Answer FIVE questions in all

SECTION A

1. Define the following terms

(i.) Electric Current (I)

(ii.) Potential difference (V)

current (ac)

2. A cell of e.m.f. 5 V is connected to

(i.) current in the circuit.

(ii.) power consumed by the circuit

(iii.) energy consumed, if the e.m.f. is connected for 22 min.

3. An electric heater has a rating of 2 kW at 220 V. Calculate the

(i.) current drawn by the heater when connected to a 220V supply.

(ii.) resistance of the heating element.

(iii.) Electromotive force (emf)

(iv.) Direct Current (dc) (v.) Alternating

a circuit of resistance R . Calculate the

2. (a) State Coulomb's law and express it mathematically

(b) A parallel plate capacitor has an effective area of 100 cm^2 , separated by a dielectric

6.4 mm thick. Its capacitance is 44 pF and it is charged to a potential difference of 16.4

V

(c) Calculate the

(i.) electric flux density in the dielectric

(ii.) electric field strength

iii. relative permittivity 0

space, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m

{ the dielectric material. assuming the permittivity in free

af Define the following terms

(i.) Magnetic flux (Φ) i.) Magnetic flux density (B)

(iii. Magnetomotive force (mmf) _ (iv.) Magnetic field strength (B)

b. A non-magnetic circuit of mean length 35 cm and cross-section 2.5 cm² is uniformly wound with 850 turns carrying 4 d.c of 4A. Calculate the

i. magnetomotive force (mmf)

ii. magnetizing force (H)

iii, magnetic flux density (B)

iv. flux (Φ)

v. reluctance (S)

SECTION B

ffs first and second law as related to DC circuit

4. a. With neat sketches. State Kircho

beorem?

b. What are the limitations of superpositior t

c. State Thevenin's theorem and draw the equivalent

Gd. Use the loop (mesh) curren! method to find the curre

circuit

nt in R of Figure Q4

10 10

Figure Q4

5V

5. a. With a suitable sketch. state the maximum power transfer theorem

h. Define the following terms as it relate to DC circuit analysis

1 Mesh 11) Node wy. Path 1) Branch v). Loop

q. Find the branch current in the circuit of Figure Q4 by using nodal analysis

d. Using Norton's Theorem. calculate the current through R in Figure QS

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. ed 14 | Rit 5)

B

Figure QS

SECTION C

6. Figure Q6 is the circuit diagram of the loads assigned to a hypothetical room. Determine the

- i. Real power (P)
- i. Watt-less power Q
- ii. Apparent Power (S)

vv. Draw the triangle of power of Figure and determine the current that flows through the circuit

600 W, 600 var

1000 VAR

ind Cap.

50 VAR Cap

eo. Vv

50 W and 150 VAR

FO VAR

Cap

tres

Figure Q6: Circuit diagram of loads assigned to a hypothetical room

7. Figure Q7 is a circuit diagram representing the power consumed by a heater and Watt-less gadgets, Compute the

- i. Current through the heater
- ii. Power and state reactive power if it is an inductance or capacitive reactance
- 1. By applying the voltage divider's rule. calculate the voltage across the capacitor

10 V 2) 3 : in case

_t

Xe

25kW

heater

Figure Q7: Circuit diagram of power consumed by a heater and a Watt-less gadget

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Paper 2 of 2 — GET 201 Applied Electricity I

Level 200 • First Semester • PastQuestion

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EXAMINATION; First Semester . COURSE CODE: GET 20]

SESSION: 26242025 oe COURSE TITLE: Applied Electricity |

DATE: 21/05/2025 Shee TIME ALLOWED: 3 Hrs.

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- ii. magnetizing force (H)
- iii. magnetic flux density (B)
- iv. flux (Φ)
- v. reluctance (S)

SECTION p

4. a. With neat sketches. State Kirchhoff's first and Second Law as related to DC
- b. What are the limitations of superposition theorem? state three
- c. State Thevenin's theorem and draw the equivalent circuit
- a. Use the loop (mesh) current method to find the current in R of Figure Q4

Figure Q4 #

- a. With a suitable sketch, state the maximum power transfer theorem
- b. Define the following terms as they relate to DC circuit:
 - i. Mesh
 - ii. Node
 - iii. Path
 - iv. Branch
- c. Find the branch current in the circuit of Figure Q4 by using nodal analysis
- d. Using Norton's Theorem, calculate the current through R in Figure Q4

) 4

9 R(t 5)

fe

/ Figure Q5

SECTION C

Q6 is the circuit diagram of the loads assigned to a hypothetical room. Determine the

- i. Real power (P)
- ii. Watt-less power (Q)
- iii. Apparent Power (S)
- iv. Draw the triangle of power of Figure and determine the current that flows through the circuit

800 W, 500 VAR

1.000 VAR |

Ind. | Cap

D VAI

50 VAR Cap.

120V

1 |

50 W and 150 VAR

70 VAR ,

I | I_0 , |

1 | Lat.t

Tie — ——*

Figure Qo: Circuit diagram of loads assigned in a hypothetical room

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